THE KESTEN–STIGUM BOUND IS TIGHT FOR THE THREE-STATE POTTS MODEL ON EVERY REGULAR TREE

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ABSTRACT. A uniformly random colour from $\{1,2,3\}$ is broadcast from the root of the infinite d-ary tree; each edge keeps its parent's colour with probability $\lambda + (1-\lambda)/3$ and otherwise moves to one of the two other colours uniformly. We prove that for every $d \geq 2$ and every $0 \leq \lambda \leq d^{-1/2}$ the colours at depth n carry asymptotically no information about the root. Together with the theorem of Kesten and Stigum this shows that the reconstruction threshold of the ferromagnetic three-state Potts channel on the d-ary tree is $\lambda = d^{-1/2}$ for every degree. Sly, and Mossel, Sly and Sohn, proved this for all degrees above a constant that was not made explicit. Our proof makes that constant explicit, $d \geq 91$, by a quantitative Gaussian approximation, and covers $2 \leq d \leq 90$ by computer-assisted arguments: an exact finite-state upgrade of the belief-propagation recursion bounds the second moment of the root posterior at a finite depth, and exact polynomial certificates exclude every nontrivial belief-propagation fixed point below that bound.

1. INTRODUCTION

Let T_d be the infinite rooted d-ary tree, $q \geq 2$ and $\lambda \in [0,1]$, and let $M_\lambda = \lambda I + (1-\lambda)J$ be the $q \times q$ Potts channel, where every entry of J is $1/q$. The broadcast process gives the root ρ a uniform colour $\sigma_\rho \in [q]$ and passes colours down each edge independently through M_λ. Write σ_{L_n} for the colours at depth n. The process is *reconstructible* if

$$\lim_{n\to\infty} \mathbb{E}\left\|\mathbb{P}(\sigma_\rho = \cdot \mid \sigma_{L_n}) - q^{-1}\mathbf{1}\right\|_{\mathrm{TV}} > 0$$

and *nonreconstructible* otherwise. Kesten and Stigum [5] showed that reconstruction holds whenever $d\lambda^2 > 1$. For $q = 2$ the bound is sharp [1, 3]. For large q it is not: Mossel [6] showed that reconstruction can hold below it, and Sly [8] proved that the bound fails to be tight for $q \geq 5$ and that it is tight for $q = 3$ once $d \geq d_0$. Mossel, Sly and Sohn [7] extended tightness to $q = 4$ and to Galton–Watson trees of large mean. In both papers d_0 comes from a central limit approximation and is not computed, and Mossel, Sly and Sohn ask whether the result holds for $q = 3$ and small d. Bhatnagar and Maneva [2] gave a computational method that bounds reconstruction thresholds on small trees; a finite computation of this kind cannot by itself reach the Kesten–Stigum bound, since at $d\lambda^2 = 1$ the information decays only polynomially in the depth.

Theorem 1.1. *Let $q = 3$. For every integer $d \geq 2$ and every $0 \leq \lambda \leq d^{-1/2}$, the broadcast process on T_d is nonreconstructible. Hence the reconstruction threshold of the ferromagnetic three-state Potts channel on the d-ary tree equals the Kesten–Stigum bound $\lambda = d^{-1/2}$ for every $d \geq 2$.*

The proof follows one plan at every degree. Let $x_n = \mathbb{E}\sum_i (p_i - 1/3)^2$, where $p = \mathbb{P}(\sigma_\rho = \cdot \mid \sigma_{L_n})$. The sequence x_n is nonincreasing, its limit x_∞ is the second moment of a colour-symmetric fixed point of the belief-propagation map (Section 2), and nonreconstruction is the statement $x_\infty = 0$. Two estimates meet:

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(1) an *exclusion radius* $r_d > 0$: no nontrivial symmetric fixed point at the critical $\lambda = d^{-1/2}$ has second moment in $(0, r_d]$ (Section 3);
(2) an *entry bound*: $x_N < r_d$ for some explicit depth N (Sections 4 and 5).

At the critical point the drift of x near zero is $-\frac{3}{4}\lambda^4 d(d-1)x^2$ to leading order ([8, (4.9)]), so the fixed-point equation leaves no room for a small nonzero solution; the work in (1) is to make the neighbourhood explicit. We do this with exact Bernstein-basis certificates for polynomial inequalities in the low moments of the fixed point. For (2) at $2 \le d \le 90$ we run the belief-propagation recursion on an information upgrade of the posterior law that is supported on a finite grid, in exact integer arithmetic; the upgrade can only increase x_n, so the computed value is an upper bound. For $d \ge 91$ we replace the computation by a Lindeberg estimate with every constant written down, comparing the recursion with the Gaussian map of Sly [8], together with a certified lower bound on the gap between that map and the identity. Monotonicity in λ reduces subcritical λ to the critical one (Section 6).

Table 2 in Section 6 lists, for each degree or band of degrees, the radius and the certified bound. The computations are in the accompanying repository and are described in Section 7.

2. The recursion and its fixed points

Throughout, $q = 3$, $u = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, and Δ_3 is the probability simplex in $\mathbb{R}^3$ with the Euclidean norm $|\cdot|$. The symmetric group S_3 acts on Δ_3 by permuting coordinates. For a Borel probability measure μ on Δ_3 write

$$x(\mu) = \int |p - u|^2 \, \mu(dp).$$

We say that μ is *symmetric* if it is S_3-invariant, and that μ is a *posterior law* if there are a random colour σ, uniform on $[3]$, and a σ-algebra $\mathcal{G}$ such that μ is the law of $\mathbb{P}(\sigma = \cdot \mid \mathcal{G})$. A posterior law has mean u, and $x(\mu) = 0$ only for the point mass δ_u.

2.1. **The recursion.** Fix $d \ge 2$ and $0 \le \lambda < 1$. For a probability measure μ on Δ_3 with mean u let $p^{(1)}, \dots, p^{(d)}$ be independent with law μ, and put

$$z_i = \prod_{j=1}^{d} \bigl(1 - \lambda + 3\lambda p_i^{(j)}\bigr) \quad (i = 1, 2, 3), \qquad S = z_1 + z_2 + z_3.$$

Define $T_\lambda \mu$ by

$$\int h \, d(T_\lambda \mu) = \mathbb{E}\Bigl[h\bigl(z/S\bigr)\,\frac{S}{3}\Bigr] \tag{1}$$

for bounded measurable h. Each factor $1-\lambda+3\lambda p_i$ has mean 1 under μ, so $\mathbb{E}z_i = 1$ and $\mathbb{E}S/3 = 1$; thus $T_\lambda\mu$ is a probability measure, and it again has mean u because $\int p_i \, d(T_\lambda\mu) = \mathbb{E}[z_i]/3 = 1/3$.

Let $P_n = \mathbb{P}(\sigma_\rho = \cdot \mid \sigma_{L_n})$ and let μ_n be its law. Thus μ_0 is the uniform law on the three vertices of Δ_3.

Lemma 2.1. *$\mu_{n+1} = T_\lambda \mu_n$ for every $n \ge 0$. Each μ_n is a symmetric posterior law.*

Proof. Let $v_1, \dots, v_d$ be the children of ρ, and let Y_j be the colours at depth $n+1$ below v_j, so that $\sigma_{L_{n+1}} = (Y_1, \dots, Y_d)$. The subtree below v_j is a copy of T_d, and σ_{v_j} is uniform; hence $Q^{(j)} = \mathbb{P}(\sigma_{v_j} = \cdot \mid Y_j)$ has law μ_n. The Y_j are not independent, since they share the root; we compute their joint law. By Bayes' rule with a uniform prior, $\mathbb{P}(Y_j \mid \sigma_{v_j} = k) = 3Q_k^{(j)}\,\mathbb{P}(Y_j)$. Summing over the colour of v_j,

$$\mathbb{P}(Y_j \mid \sigma_\rho = i) = \sum_k M_\lambda(i,k)\, 3Q_k^{(j)}\,\mathbb{P}(Y_j) = \bigl(1 - \lambda + 3\lambda Q_i^{(j)}\bigr)\mathbb{P}(Y_j).$$

Since the subtrees are conditionally independent given σ_ρ, $\mathbb{P}(Y_1,\dots,Y_d \mid \sigma_\rho = i) = z_i \prod_j \mathbb{P}(Y_j)$ with z built from the $Q^{(j)}$. The posterior of the root is therefore z/S, and, averaging over the uniform root, the joint law of $(Y_1,\dots,Y_d)$ has density $\frac{1}{3}\sum_i z_i = S/3$ with respect to the product of its marginals. Since z is a function of $(Q^{(1)},\dots,Q^{(d)})$, the vector $(Q^{(1)},\dots,Q^{(d)})$ has density $S/3$ with respect to $\mu_n^{\otimes d}$, which is (1). Symmetry passes from μ_n to μ_{n+1} because M_λ commutes with permutations of the colours, and μ_n is a posterior law by definition. □

2.2. **The tail limit.** Let $\mathcal{F}_n$ be the σ-algebra generated by all colours at depth at least n. These σ-algebras decrease. Given σ_{L_n}, the root is independent of the colours below depth n, so $P_n = \mathbb{P}(\sigma_\rho = \cdot \mid \mathcal{F}_n)$. Put $x_n = x(\mu_n) = \mathbb{E}|P_n - u|^2$ and $\mathcal{F}_\infty = \bigcap_n \mathcal{F}_n$.

Lemma 2.2. *The sequence x_n is nonincreasing. The posteriors P_n converge almost surely and in L^2 to $P_\infty = \mathbb{P}(\sigma_\rho = \cdot \mid \mathcal{F}_\infty)$, whose law μ_∞ is a symmetric posterior law with $T_\lambda\mu_\infty = \mu_\infty$ and $x(\mu_\infty) = \lim_n x_n \le x_N$ for every N.*

Proof. (P_n) is a reverse martingale for the decreasing filtration $(\mathcal{F}_n)$, so $P_{n+1} = \mathbb{E}[P_n \mid \mathcal{F}_{n+1}]$, and Jensen's inequality gives $x_{n+1} \le x_n$. Lévy's downward theorem gives convergence almost surely, and in L^2 by boundedness, to P_∞. Hence $\mu_n \to \mu_\infty$ weakly and $x_n \to x(\mu_\infty)$. Symmetry passes to the limit, and μ_∞ is a posterior law by construction.

For the fixed-point identity, each factor in z is at least $1-\lambda > 0$, so $S \ge 3(1-\lambda)^d$, and the integrand in (1) is a bounded continuous function of $(p^{(1)},\dots,p^{(d)}) \in \Delta_3^d$ when h is bounded and continuous. Weak convergence $\mu_n \to \mu_\infty$ implies weak convergence of the product laws $\mu_n^{\otimes d} \to \mu_\infty^{\otimes d}$, hence $T_\lambda\mu_n \to T_\lambda\mu_\infty$ weakly. By Lemma 2.1, $T_\lambda\mu_n = \mu_{n+1} \to \mu_\infty$, so $T_\lambda\mu_\infty = \mu_\infty$. □

Proposition 2.3. *Let $0 < \lambda < 1$ and $r > 0$. Suppose that every symmetric posterior law μ with $T_\lambda\mu = \mu$ and $x(\mu) \le r$ equals δ_u, and that $x_N \le r$ for some N. Then the broadcast process with parameters (d,λ) is nonreconstructible.*

Proof. By Lemma 2.2, μ_∞ satisfies the hypotheses, so $\mu_\infty = \delta_u$ and $x_n \to 0$. By Cauchy–Schwarz in $\mathbb{R}^3$ and Jensen, $\mathbb{E}\|P_n - u\|_{\mathrm{TV}} = \frac{1}{2}\mathbb{E}\sum_i |P_n(i) - \frac{1}{3}| \le \frac{\sqrt{3}}{2}\sqrt{x_n} \to 0$. □

2.3. **Comparison of experiments.** Write $x_n(d,\lambda)$ for x_n when the parameters need to be named.

Lemma 2.4. *If $2 \le d \le D$ and $0 \le \lambda \le \eta < 1$, then $x_n(d,\lambda) \le x_n(D,\eta)$ for every n.*

Proof. Both changes are garblings of the observation that do not depend on the root. If $Y = g(X,U)$ with U independent of (σ_ρ, X), then $\mathbb{P}(\sigma_\rho = \cdot \mid Y) = \mathbb{E}[\mathbb{P}(\sigma_\rho = \cdot \mid X) \mid Y]$, and Jensen's inequality for the convex function $p \mapsto |p-u|^2$ shows that the second moment can only decrease. For the degree, T_d is a subtree of T_D containing the root, and the depth-n colours of T_d are a subset of those of T_D. For the strength, if $\eta > 0$ then

$$M_\lambda = \tfrac{\lambda}{\eta} M_\eta + \big(1 - \tfrac{\lambda}{\eta}\big) J,$$

and an edge of strength λ is an edge of strength η with probability λ/η and otherwise gives the child a uniform colour independent of its parent. Mark each edge independently with probability $1-\lambda/\eta$. On the vertices joined to the root by unmarked paths, keep the colours of the strength-η process; below every marked edge with no marked edge above it, put a fresh copy of the strength-λ process started from a uniform colour. The result is the strength-λ process, and its depth-n colours are a function of the depth-n colours of the strength-η process together with the marks and the fresh copies, which are independent of the root and of that process. □

By Lemma 2.4 with $D = d$, nonreconstruction at $\lambda = d^{-1/2}$ implies it for every smaller λ, and an upper bound on $x_N(D,\eta)$ with $D \ge d$ and $\eta \ge d^{-1/2}$ is an upper bound on $x_N(d, d^{-1/2})$. The second use lets one computation serve a band of degrees (Section 4).

3. Exclusion near the uniform law

In this section $\lambda = d^{-1/2}$ unless stated otherwise. We prove that a nontrivial symmetric fixed point cannot have small second moment, with the radii $r_2 = \frac{23}{2000}$, $r_3 = \frac{3}{100}$ and, for $d \ge 4$,

$$r_d = \max\Big\{\tfrac{5}{36}\big(1 - d^{-1/2}\big)^2,\ \varrho_d\Big\}, \qquad \varrho_d = \begin{cases} 0, & d = 4, \\ \frac{217}{4500}, & 5 \le d \le 19, \\ \frac{257}{3000}, & 20 \le d \le 79, \\ \frac{179}{1500}, & d \ge 80. \end{cases} \tag{2}$$

3.1. **Fusion and moments.** For $p \in \Delta_3$ write $a(p) = 3p - \mathbf{1}$, a vector with coordinate sum zero and $|a|^2 \le 6$. For a symmetric probability measure μ on Δ_3 put

$$M(\mu) = \int |a|^2 d\mu, \qquad U(\mu) = \int |a|^4 d\mu, \qquad T(\mu) = \int \textstyle\sum_i a_i^3 \, d\mu,$$

so that $x(\mu) = M(\mu)/9$ and $0 \le M \le 6$. For symmetric μ, ν define the *fusion* $\mu \circledast \nu$ by

$$\int h \, d(\mu \circledast \nu) = \mathbb{E}\Big[h\Big(\frac{p \odot r}{\langle p, r\rangle}\Big)\, 3\langle p, r\rangle\Big], \qquad p \sim \mu,\ r \sim \nu \text{ independent},$$

where $\odot$ is the coordinatewise product. If μ and ν are the laws of the posteriors of a uniform colour σ given two observations that are conditionally independent given σ, then, exactly as in the proof of Lemma 2.1, $\mu \circledast \nu$ is the law of the posterior given both. Hence $\circledast$ is commutative and associative on symmetric posterior laws, and $M(\mu \circledast \nu) \ge M(\mu)$ by the argument of Lemma 2.4. Let $E_\lambda \mu$ be the law of $u + \lambda(p - u)$ for $p \sim \mu$; it is the law of the posterior of ρ given the subtree of one child whose own posterior has law μ. Since $1 - \lambda + 3\lambda p_i = 3(u + \lambda(p - u))_i$,

$$T_\lambda \mu = (E_\lambda \mu)^{\circledast d}, \qquad M(E_\lambda \mu) = \lambda^2 M(\mu), \quad U(E_\lambda \mu) = \lambda^4 U(\mu), \quad T(E_\lambda \mu) = \lambda^3 T(\mu).$$

Every p in the support of $E_\lambda \mu$ has $p_i \ge (1 - \lambda)/3$.

For $p, r \in \Delta_3$ put $\alpha = a(p)$, $\beta = a(r)$, $c = \langle \alpha, \beta\rangle$, $S = 3 + c = 9\langle p, r\rangle$ and

$$v = \alpha + \beta + \alpha \odot \beta - \tfrac{c}{3}\mathbf{1}, \qquad N = |v|^2.$$

A direct computation gives $a\big(p \odot r/\langle p, r\rangle\big) = 3v/S$, so the fused vector has a-norm $9N/S^2$ and weight $S/3$.

Lemma 3.1. *Let μ, ν be symmetric probability measures on Δ_3 with moments (m, V, T_α) and (n, W, T_β) respectively, and suppose $S \ge s > 0$ on the support of $\mu \otimes \nu$. Then*

$$M(\mu \circledast \nu) = m + n - \tfrac{mn}{6} - \tfrac{1}{3}(nT_\alpha + mT_\beta) + R, \qquad 0 \le R \le \frac{\mathbb{E}[Nc^2]}{3s},$$

$$\mathbb{E}[Nc^2] = \tfrac{1}{2}(nV + mW) + 3T_\alpha T_\beta + \tfrac{1}{2}(VT_\beta + T_\alpha W) + \tfrac{1}{12}VW.$$

Proof. By the remark before the lemma, $M(\mu \circledast \nu) = \mathbb{E}[(9N/S^2)(S/3)] = 3\mathbb{E}[N/S]$. Since $(\frac{1}{3} - \frac{c}{9})S = 1 - \frac{c^2}{9}$, we have $1/S = \frac{1}{3} - \frac{c}{9} + \frac{c^2}{9S}$, hence $M(\mu \circledast \nu) = \mathbb{E}N - \frac{1}{3}\mathbb{E}[Nc] + R$ with $R = \mathbb{E}[Nc^2/(3S)]$, which lies between 0 and $\mathbb{E}[Nc^2]/(3s)$. A symmetric law of a zero-sum vector α with $\mathbb{E}|\alpha|^2 = m$, $\mathbb{E}|\alpha|^4 = V$, $\mathbb{E}\sum \alpha_i^3 = T_\alpha$ has $\mathbb{E}\alpha = 0$ and, for distinct indices i, j, k,

$$\mathbb{E}\alpha_i^2 = \tfrac{m}{3},\ \mathbb{E}\alpha_i\alpha_j = -\tfrac{m}{6}; \quad \mathbb{E}\alpha_i^3 = \tfrac{T_\alpha}{3},\ \mathbb{E}\alpha_i^2\alpha_j = -\tfrac{T_\alpha}{6},\ \mathbb{E}\alpha_1\alpha_2\alpha_3 = \tfrac{T_\alpha}{3};$$

$$\mathbb{E}\alpha_i^4 = \tfrac{V}{6},\ \mathbb{E}\alpha_i^3\alpha_j = -\tfrac{V}{12},\ \mathbb{E}\alpha_i^2\alpha_j^2 = \tfrac{V}{12},\ \mathbb{E}\alpha_i^2\alpha_j\alpha_k = 0.$$

(The second and fourth tables follow from symmetry and $\sum_i \alpha_i = 0$; for the fourth, also $\sum_i \alpha_i^4 = \frac{1}{2}|\alpha|^4$ for zero-sum triples.) Expanding N, Nc and Nc^2 as polynomials in α, β and contracting with these tables gives $\mathbb{E}N = m + n + \frac{mn}{6}$, $\mathbb{E}[Nc] = mn + nT_\alpha + mT_\beta$ and the stated formula for $\mathbb{E}[Nc^2]$. $\square$

3.2. **Three moment inequalities.** Fix $\eta = 2887/5000$ and $\kappa = 2(99/140)^4 < \frac{1}{2} + 10^{-4}$.

Lemma 3.2. *Let μ, ν be symmetric posterior laws with moments (M_μ, U_μ, T_μ) and (M_ν, U_ν, T_ν).*

(CF) $T(\mu \circledast \nu) \geq (1 - \frac{M_\nu}{6})T_\mu + (1 - \frac{M_\mu}{6})T_\nu$.
(F) *If ν is supported on $\{r : r_i \geq (1-\eta)/3\}$, then $U(\mu \circledast \nu) \leq U_\mu + U_\nu + 5M_\mu M_\nu$.*
(C) *Under the same hypothesis, $|T(\mu \circledast \nu) - T_\mu - T_\nu| \leq \frac{5}{3} M_\mu M_\nu$.*
(F1) *If $0 < \lambda \leq 1/\sqrt{2}$, then $\lambda^4 U\big((E_\lambda \mu)^{\circledast 2}\big) \leq \kappa \lambda^4 U_\mu + \frac{3}{2}\lambda^4 M_\mu^2$.*

Proof. Each inequality is bilinear in the pair (μ, ν) once the terms that involve one law only are multiplied by the total mass of the other, and the fusion is bilinear by its definition. A symmetric law is a mixture of uniform laws on S_3-orbits, so it suffices to prove each inequality when μ and ν are uniform on the orbits of two points. Write the sorted points as

$$p(t, y) = \big(\tfrac{1-t}{3},\ \tfrac{1-t}{3} + \tfrac{ty}{2},\ \tfrac{1-t}{3} + t(1 - \tfrac{y}{2})\big), \qquad 0 \leq t, y \leq 1,$$

and similarly $r = p(s, z)$. The fused law is then the average over the six relative permutations π of r, the term for π carrying weight $S_\pi/18$ and output a-vector $3v_\pi/S_\pi$. After clearing the positive denominators S_π (in (CF) the terms with $S_\pi = 0$ carry no weight), each inequality becomes the nonnegativity of an explicit polynomial with integer coefficients on a product of intervals in (t, y, s, z). We prove this nonnegativity in Section 7 by dividing out the monomial factors that vanish on the boundary and expanding the quotient in a Bernstein basis, all of whose coefficients are nonnegative. For (F1) the reduction is first made at $\lambda = 99/140$; the case of smaller λ follows by Lemma 2.4 applied to the two-child tree, since the right side is then computed with the same scale. □

Corollary 3.3. *If $d\lambda^3 < 1$, then every symmetric posterior law μ with $T_\lambda \mu = \mu$ has $T(\mu) \geq 0$.*

Proof. Let $T = T(\mu)$ and let T_j be the cubic moment of $(E_\lambda \mu)^{\circledast j}$, so $T_1 = \lambda^3 T$ and $T_d = T$. Suppose $T < 0$. Since the coefficients in (CF) lie in $[0, 1]$, (CF) gives $T_{j+1} \geq \min(T_j, 0) + \lambda^3 T$, hence $T_j \geq j\lambda^3 T$ for all j. At $j = d$ this reads $(1 - d\lambda^3)T \geq 0$, a contradiction. □

3.3. **Degree two.**

Proposition 3.4. *Let $d = 2$ and $\lambda = 2^{-1/2}$. A symmetric posterior law μ with $T_\lambda \mu = \mu$ and $x(\mu) \leq r_2$ equals δ_u.*

Proof. Let m, V, t be the moments of the child law $E_\lambda \mu$, so $m = M(\mu)/2 = \frac{9}{2} x(\mu)$, $V = U(\mu)/4$ and $\mu = (E_\lambda \mu)^{\circledast 2}$. By (F1) at the fixed point, $V \leq \kappa V + \frac{3}{2} m^2$, so $V \leq C m^2$ with $C = \frac{3001}{1000}$. For a zero-sum vector, $|\sum_i \alpha_i^3| \leq |\alpha|^3/\sqrt{6}$, so by Cauchy–Schwarz $|t| \leq \sqrt{mV/6} \leq \theta m^{3/2}$ with $\theta = \frac{177}{250} \geq \sqrt{C/6}$. For two child vectors $c \geq -3\lambda^2$, so $S \geq 3/2$. Lemma 3.1 with both laws equal to $E_\lambda \mu$ and the identity $M(\mu) = 2m$ give

$$\tfrac{m^2}{6} \leq \tfrac{2}{3} m|t| + \tfrac{2}{9}\big(mV + 3t^2 + V|t| + \tfrac{1}{12} V^2\big).$$

If $m > 0$, divide by m^2 and insert the bounds above:

$$\tfrac{1}{6} \leq \tfrac{2}{3}\theta\sqrt{m} + \tfrac{2}{9}\big(\tfrac{3}{2} C m + C\theta m^{3/2} + \tfrac{1}{12} C^2 m^2\big).$$

For $x(\mu) \leq \frac{23}{2000}$ we have $m \leq \frac{207}{4000}$ and $\sqrt{m} \leq \frac{91}{400}$, and the right side is at most $0.16516 < \frac{1}{6}$ (exact rational arithmetic). Hence $m = 0$. □

3.4. **Degree at least three.** Let $d \geq 3$, $\lambda = d^{-1/2}$, and let μ be a nontrivial symmetric posterior fixed point with moments M, U, T. Let $\mu_j = (E_\lambda \mu)^{\circledast j}$, with moments M_j, U_j, T_j, so $\mu_d = \mu$ and $M_j \leq M$. The child law has moments M/d, U/d^2, $\lambda T/d$, and its support satisfies $r_i \geq (1-\lambda)/3$ with $\lambda < \eta$. By Corollary 3.3 (note $d\lambda^3 = \lambda < 1$) and (CF), $T \geq 0$ and $T_j \geq 0$ for all j. For $2 \leq k \leq d$ set

$$g_k = \frac{k(k-1)}{12d^2}, \qquad Z = \frac{k(k-1)}{2} + d(d-k), \qquad C_U = \frac{5Z}{d(d-1)}, \qquad C_T = \frac{C_U(1+\lambda)}{3},$$

and let $A_k, B_k \geq 0$ be the closed forms

$$A_k = A_2 + \frac{S_2}{72d^3} + \frac{K(C_U S_1 + \frac{5}{4}S_2)}{d^3}, \qquad B_k = B_2 + K\Big[(3t_e + \tfrac{v_e}{2})\Sigma_t + (\tfrac{t_e}{2} + \tfrac{v_e}{12})\Sigma_u\Big],$$

where $S_1 = \frac{k(k-1)}{2} - 1$, $S_2 = \frac{k(k-1)(k-2)}{3}$, $K = \frac{1}{9(1-\lambda)}$, $K' = \frac{d}{9(d-1)}$, $v_e = C_U/d^2$, $t_e = \lambda C_T/d$, $\Sigma_u = (C_U S_1 + \frac{5}{2}S_2)/d^2$, $\Sigma_t = \lambda C_T S_1/d + \frac{5}{6}S_2/d^2$, $A_2 = K' C_U/d^3$ and $B_2 = K'\big[3C_T^2/d^3 + \lambda C_U C_T/d^3 + C_U^2/(12d^4)\big]$.

Lemma 3.5. *Let $0 < M_* \leq 6$ and $0 < \delta < 1$, and suppose that*

$$(1-\delta)g_k - M_* A_k - M_*^2 B_k \geq 0 \qquad (2 \leq k \leq d). \tag{3}$$

Then no nontrivial symmetric posterior fixed point has $M \leq M_*$.

Proof. Suppose $0 < M \leq M_*$. We show by induction on $k = 2, \ldots, d$ that

$$M_k \leq \tfrac{k}{d}M - g_k M^2 + A_k M^3 + B_k M^4, \tag{4}$$

and hence, by (3), $M_k \leq \frac{k}{d}M - \delta g_k M^2 < \frac{k}{d}M$. At $k = d$ this contradicts $M_d = M$.

Assume $M_j \leq jM/d$ for $1 \leq j < k$ (an equality at $j = 1$). Together with $M_j \leq M$ for $j \geq k$ this gives $\sum_{j=1}^{d-1} M_j \leq M\big[\frac{k(k-1)}{2d} + d - k\big] = MZ/d$. Applying (F) and (C) to $\mu_{j+1} = \mu_j \circledast E_\lambda \mu$ and telescoping from $j = 1$ to $j = d-1$,

$$\big(1 - \tfrac{1}{d}\big)U \leq \tfrac{5M}{d}\sum_{j<d} M_j, \qquad (1-\lambda)T \leq \tfrac{5M}{3d}\sum_{j<d} M_j,$$

so, using $(1-\lambda)(1+\lambda) = (d-1)/d$, $U \leq C_U M^2$ and $0 \leq T \leq C_T M^2$. Telescoping the same inequalities only up to $j < k$ gives $U_j \leq u_j M^2$ and $T_j \leq t_j M^2$ with $u_j = [jC_U + \frac{5}{2}j(j-1)]/d^2$ and $t_j = j\lambda C_T/d + \frac{5}{6}j(j-1)/d^2$.

For $k = 2$, apply Lemma 3.1 to two child laws: their vectors satisfy $c \geq -3\lambda^2$, so $S \geq 3(d-1)/d$, and the cubic term is nonpositive since $T \geq 0$. This gives (4) at $k = 2$. For the step from j to $j+1 < k+1$, apply Lemma 3.1 with left law μ_j and right law $E_\lambda \mu$: here $S \geq 3(1-\lambda)$ because every right coordinate is at least $(1-\lambda)/3$, the cubic term is nonpositive, and the bounds above give

$$R \leq K\Big[\tfrac{u_j}{2d} + \tfrac{jC_U}{2d^3}\Big]M^3 + K\Big[3t_j t_e + \tfrac{1}{2}(u_j t_e + t_j v_e) + \tfrac{1}{12}u_j v_e\Big]M^4.$$

The main term is $(1 - \frac{M}{6d})M_j + \frac{M}{d}$, with a positive multiplier since $M \leq 6$. Inserting (4) for M_j and dropping the nonpositive products of the multiplier's $-M/(6d)$ with the cubic and quartic terms, the quadratic coefficient becomes $g_j + j/(6d^2) = g_{j+1}$, and the cubic and quartic coefficients grow by exactly the amounts whose sums over j are the closed forms A_k, B_k above. □

Proposition 3.6. *Let $d \geq 3$ and $\lambda = d^{-1/2}$. A symmetric posterior law μ with $T_\lambda \mu = \mu$ and $x(\mu) \leq r_d$ equals δ_u.*

Proof. We verify (3) for all integers $d \geq 3$ and $2 \leq k \leq d$ at once. Substitute $\ell = \lambda$, $d = \ell^{-2}$, $k = 2 + \rho(d-2)$ with $\rho \in [0,1]$, and $\ell = Lz$ with $z \in [0,1]$. After multiplication by a positive factor $D(\ell)$ the left side of (3) becomes a polynomial in (z, ρ), and every integer pair (d,k) with $d^{-1/2} \leq L$ corresponds to a point of $[0,1]^2$. The following choices are used.

degrees	M_*	$x = M_*/9$	δ	L
$d \geq 3$	$27/100$	$3/100$	$1/100$	$\eta = 2887/5000$
$d \geq 4$	$\frac{5}{4}(1-\ell)^2$	$\frac{5}{36}(1-\lambda)^2$	$1/1000$	$1/2$
$d \geq 5$	$217/500$	$217/4500$	$1/1000$	$559017/1250000$
$d \geq 20$	$771/1000$	$257/3000$	$1/1000$	$559017/2500000$
$d \geq 80$	$537/500$	$179/1500$	$1/1000$	$559017/5000000$

In each case $L \geq d^{-1/2}$ at the smallest degree of the row, and the Bernstein coefficients of the polynomial on $[0,1]^2$ are nonnegative (Section 7). Hence (3) holds and Lemma 3.5 applies with $x = M/9$. For $d \geq 4$ the curve dominates the constant $3/100$, since $(1 - d^{-1/2})^2 \geq \frac{1}{4} > \frac{27}{125}$, so r_d in (2) is the largest radius available. □

4. Certified upper envelopes

By Lemma 2.2, an entry bound $x_N(d, d^{-1/2}) < r_d$ is needed at one depth N only. We obtain it from a finite computation whose output is more informative than the true posterior law at every depth, so that its second moment is an upper bound. The method is that of Bhatnagar and Maneva [2]. We state the version used here in full, because the certificates depend on its details: observations of probability zero, the discretization and the rounding of masses are all part of the argument.

4.1. **Two upgrades.** Let μ, ν be symmetric posterior laws. Say that ν is *more informative* than μ if there are observations X and Y of a uniform colour σ, with posterior laws ν and μ, such that $Y = g(X, U)$ with U independent of (σ, X). As in the proof of Lemma 2.4, this implies $x(\nu) \geq x(\mu)$. The relation is preserved by E_λ and by fusion with any third law, since the garbling can be applied to one observation and the others left alone.

Fix an integer $K \geq 1$ and let $G_K = \{v/K : v \in \mathbb{Z}^3_{\geq 0},\ v_1 + v_2 + v_3 = K\}$. For $p \in \Delta_3$ write $Kp = f + r$ with f integral and $0 \leq r_i < 1$, so that $r_1 + r_2 + r_3 \in \{0, 1, 2\}$. The *split* of p is the point p itself if this sum is 0; the points $(f + e_i)/K$ with weights r_i if it is 1; and the points $(f + \mathbf{1} - e_i)/K$ with weights $1 - r_i$ if it is 2. In each case the weights are nonnegative, sum to one and average to p, and the split commutes with permutations of the colours.

Lemma 4.1. *Let μ be a symmetric posterior law and let μ' be the law obtained by replacing each p by its split. Then μ' is a symmetric posterior law on G_K, more informative than μ.*

Proof. Let X be an observation with posterior $p = \sum_j w_j q_j$, the split of p. Given X and $\sigma = i$ with $p_i > 0$, draw a label j with probability $w_j (q_j)_i / p_i$; these probabilities sum to one. If $p_i = 0$ then $(q_j)_i = 0$ whenever $w_j > 0$, the pair $(X, \sigma = i)$ has probability zero, and the kernel may be chosen arbitrarily there. By Bayes' rule the posterior given (X, j) is q_j, and given X the label equals j with probability w_j. Forgetting the label recovers X. The posterior law of (X, j) is μ'. □

A symmetric law on G_K is determined by the masses of its S_3-orbits; for $K = 96$ there are 817 orbits, of sizes 1, 3 and 6. Fix $S = 2^b$. The *rounding* of such a law replaces each orbit mass w by $\lfloor Sw \rfloor / S$ and adds the total removed mass to the orbit $\{e_1, e_2, e_3\}$.

Lemma 4.2. *The rounding of a symmetric posterior law on G_K is a symmetric posterior law, more informative than it.*

Proof. For a posterior law, $\mathbb{P}(X = p \mid \sigma = i) = 3p_i\,\mathbb{P}(X = p)$. The points of an orbit average to u, so the orbit containing the observation is independent of σ. On an orbit of mass w, keep the observation with probability $\lfloor Sw \rfloor/(Sw)$, independently of σ and of the point within the orbit, and otherwise reveal σ in addition. The revealed observations have posterior e_σ, and σ is uniform on them because the kept fraction does not depend on σ. The refined observation has the rounded law, and discarding the revelation recovers the original observation. □

Write Π for a split followed by a rounding. By Lemmas 4.1 and 4.2, $\Pi\mu$ is more informative than μ.

4.2. **The computed laws.** Fix $D \geq 2$, a rational $\eta = A/B \in (0,1)$ in lowest terms, K divisible by 3, and $S = 2^b$. Let ν_0 be the uniform law on $\{e_1, e_2, e_3\}$. Given $\nu = \nu_n$, the law ν_{n+1} is computed in one of two ways.

Sequential fusion ($D \leq 5$). Put $P_2 = \Pi\big(E_\eta\nu \circledast E_\eta\nu\big)$ and $P_{j+1} = \Pi\big(P_j \circledast E_\eta\nu\big)$ for $2 \leq j < D$, and set $\nu_{n+1} = P_D$. The shrunken law $E_\eta\nu$ enters each fusion exactly and is not split; every factor is a new copy of the child law ν, never the partial product P_j.

Binary fusion ($D \geq 6$). Put $\beta = \Pi(E_\eta\nu)$, $\beta_0 = \beta$ and $\beta_{k+1} = \Pi(\beta_k \circledast \beta_k)$, where the two factors are independent copies. If $D = 2^{k_1} + \cdots + 2^{k_s}$ with distinct k_t, the laws $\beta_{k_1}, \ldots, \beta_{k_s}$ are fused one at a time, with Π applied after each fusion, and the result is ν_{n+1}.

Proposition 4.3. *For every $n \geq 0$, ν_n is more informative than $\mu_n(D,\eta)$, and hence $x(\nu_n) \geq x_n(D,\eta)$.*

Proof. At $n = 0$ the two laws are equal. Suppose ν_n is more informative than $\mu_n = \mu_n(D,\eta)$. By Lemma 2.1, $\mu_{n+1} = (E_\eta\mu_n)^{\circledast D}$, and since fusion is associative and commutative on posterior laws, the right side may be grouped as in either scheme. In that grouping replace each factor $E_\eta\mu_n$ by $E_\eta\nu_n$, one at a time, and apply Π where the scheme applies it. Each replacement and each application of Π can only make the law more informative, so ν_{n+1} is more informative than μ_{n+1}. The last assertion follows from the first, as in Lemma 2.4. □

No closeness of ν_n to μ_n is used: every step can only add information, and the inequality holds however large the losses from the grid and the rounding are. When two laws on G_K are fused, pairs with $\langle p, r\rangle = 0$ have probability zero and are dropped before normalizing.

4.3. **Exact arithmetic.** Masses are stored as integers summing to S, and every transition between orbits as a vector of nonnegative integers with a common denominator. A grid point v/K shrunk by E_η has coordinates $((B-A)K + 3Av_i)/(3BK)$, so the split of a single shrunken point has denominator $3B$. For a fusion, fix one orientation of the left orbit and average over the $o \in \{1, 3, 6\}$ orientations of the right orbit; by simultaneous permutation of the colours this gives the full grouped transition. If the likelihood products are z_i with $C = z_1 + z_2 + z_3$, the split weights of z/C have denominator C, the pair has mass proportional to C, and C cancels. With the factor $36/o$, which is an integer, the transitions become integer vectors with row sum $108K^2$ for the fusion of two grid laws, $108(BK)^2$ for the first stage of sequential fusion and $108BK^2$ for its later stages.

A new orbit mass is the integer quotient of an accumulator $\sum m_i m_j t_{ij}$ by S times the row sum. All summands are nonnegative, so each accumulator is at most S^2 times the row sum. This is below 2^{128} in every row of Proposition 4.4 except $d = 2$, where the computation uses arbitrary-precision integers; the single-law step for binary fusion stays below $S \cdot 3B < 2^{64}$. The integer quotient is exactly the rounding of Lemma 4.2, and the removed units go to the vertex orbit. Finally

$$x(\nu_n) = \frac{1}{S}\sum_O m_O\,\frac{3|v_O|^2 - K^2}{3K^2},$$

where $v_O \in \mathbb{Z}^3$ is any point of the orbit O. No floating-point number enters a coefficient, a mass or a bound. The tables are checked as described in Section 7.

4.4. **The entry bounds.**

Proposition 4.4. *For each row of Table 1, each degree d in the range and each $n \geq n_0$,*

$$x_n\big(d, d^{-1/2}\big) \leq x(\nu_{n_0}) < r_d,$$

where ν_{n_0} is computed with the listed D and η, $K = 96$ and $S = 2^b$.

Table 1: Entry bounds. The column $x(\nu_{n_0})$ is the exact rational value rounded up to five places, and the last column is r_d at the smallest degree of the range.

d	D	η	n_0	b	$x(\nu_{n_0}) \leq$	r_d
2	2	707107/1000000	500	40	0.01142	0.01150
3	3	2887/5000	106	40	0.01847	0.03000
4	4	1/2	82	40	0.02108	0.03472
5	5	161/360	25	32	0.04689	0.04822
6	6	89/218	26	32	0.04765	0.04863
7	7	127/336	22	32	0.05348	0.05374
8	8	99/280	20	32	0.05732	0.05804
9	9	1/3	18	32	0.06158	0.06173
10	10	37/117	17	32	0.06472	0.06494
11	11	79/262	17	32	0.06510	0.06776
12	12	97/336	16	32	0.06822	0.07028
13	13	38/137	15	32	0.07172	0.07253
14	14	89/333	15	32	0.07205	0.07457
15	15	55/213	14	32	0.07595	0.07643
16	16	1/4	14	32	0.07601	0.07812
17	17	65/268	14	32	0.07676	0.07969
18	18	33/140	13	32	0.08104	0.08113
19	19	53/231	13	32	0.08147	0.08247
20–21	21	53/237	21	32	0.08481	0.08567
22–23	23	71/333	24	32	0.08552	0.08598
24–25	25	69/338	21	32	0.08746	0.08797
26–27	27	61/311	19	32	0.08937	0.08975
28–29	29	55/291	18	32	0.09042	0.09135
30–31	31	65/356	17	32	0.09180	0.09280
32–33	33	32/181	16	32	0.09384	0.09412
34–35	35	59/344	16	32	0.09364	0.09534
36–37	37	1/6	15	32	0.09570	0.09645
38–39	39	55/339	15	32	0.09609	0.09748
40–41	41	37/234	15	32	0.09607	0.09844
42–43	43	25/162	14	32	0.09904	0.09933
44–45	45	49/325	14	32	0.09922	0.10017
46–47	47	32/217	14	32	0.09948	0.10095
48–49	49	41/284	14	32	0.09976	0.10169
50–52	52	43/304	19	32	0.10125	0.10238
53–55	55	25/182	18	32	0.10238	0.10335
56–58	58	29/217	17	32	0.10398	0.10425
59–61	61	25/192	17	32	0.10418	0.10508
62–64	64	39/307	17	32	0.10448	0.10585

d	D	η	n_0	b	$x(\nu_{n_0}) \le$	r_d
65–67	67	33/266	16	32	0.10634	0.10657
68–70	70	41/338	16	32	0.10678	0.10725
71–73	73	40/337	16	32	0.10694	0.10788
74–76	76	5/43	16	32	0.10738	0.10847
77–79	79	31/272	16	32	0.10758	0.10904
80–83	83	35/313	66	32	0.10487	0.11933
84–88	88	37/339	19	32	0.11856	0.11933
89–90	90	529999/5000000	11	32	0.11761	0.11933

Proof. In each row D is the largest degree of the range and $\eta^2 d \ge 1$ at its smallest degree, so $\eta \ge d^{-1/2}$ for every d in the range. By Lemma 2.2, Lemma 2.4 and Proposition 4.3, for $n \ge n_0$,

$$x_n(d, d^{-1/2}) \le x_{n_0}(d, d^{-1/2}) \le x_{n_0}(D, \eta) \le x(\nu_{n_0}).$$

The inequality $x(\nu_{n_0}) < r_d$ is checked in exact arithmetic at every degree of the range; against the curve in (2) it is checked by squaring. Degrees $2 \le d \le 5$ use sequential fusion and $d \ge 6$ binary fusion. □

The rows are tight in places. At $d = 2$ the bound is 0.011415 against $r_2 = 0.0115$; at $d = 9$ and $d = 18$ the margins are about 1.5×10^{-4} and 1.0×10^{-4}.

5. Large degrees

For $d \ge 91$ we do not compute an entry bound. Instead we show that a fixed point with $T \ge 0$ cannot have second moment $x \ge \frac{1}{10}$; together with Proposition 3.6 this excludes every nontrivial fixed point. The argument compares one step of the recursion with the Gaussian map of Sly [8, (4.7)], by replacing the branches one at a time, in the manner of the Lindeberg argument of [8, Lemma 4.3] and [7, Appendix A]. The new points are explicit constants, and a cancellation of the third-order terms that uses a second observable.

5.1. **The Gaussian experiment.** Let $P = I - \frac{1}{3}\mathbf{1}\mathbf{1}^{\mathsf{T}}$ and let $F(z) = \mathrm{softmax}(z)$, so $F_i(z) = e^{z_i}/\sum_j e^{z_j}$. For $s \ge 0$ let Z be a random vector which, given $\sigma = i$, is Gaussian with mean $\frac{9s}{2} P e_i$ and covariance $\frac{9s}{2} P$, with σ uniform. The means have equal norms in the metric of the covariance, so the likelihood ratios are $e^{Z_i - Z_j}$ and $F(Z)$ is the posterior of σ given Z. Put

$$g_G(s) = \mathbb{E}\big[F_1(Z) \mid \sigma = 1\big] - \tfrac{1}{3}, \qquad U_G(s) = \mathbb{E}\big[|3F(Z) - \mathbf{1}|^4 \mid \sigma = 1\big].$$

By calibration $g_G(s)$ is the second moment x of the posterior law of Z. This is the map of [8, (4.7)] for $q = 3$ in our normalization: there the mean is $s(\frac{3}{2}, -3, -3)$ before centring and the covariance has diagonal 3 and off-diagonal $-\frac{3}{2}$, which after adding $\frac{3s}{2}$ to every logit is the law above.

Lemma 5.1. *$s - g_G(s) \ge \max\{\frac{9}{20}s^2, \frac{13}{200}s\} > 0$ for $\frac{1}{10} \le s \le \frac{2}{3}$.*

Proof. For $0 < a \le b$ the experiment at a is a garbling of the one at b: if Y_b has the law at b, then $\sqrt{a/b}\, Y_b + \sqrt{1 - a/b}\, Y'$, with Y' an independent centred copy of the noise, has the law at a. Hence g_G is nondecreasing, as in Lemma 2.4, and for $l \le s \le r$

$$s - g_G(s) \ge l - g_G(r). \tag{5}$$

In coordinates, with $\xi = (\xi_1, \xi_2)$ standard Gaussian,

$$g_G(s) = \mathbb{E} f_s(\xi) - \tfrac{1}{3}, \qquad f_s(\xi) = \Big[1 + 2\exp\big(-\tfrac{9s}{2} - \tfrac{3}{2}\sqrt{3s}\,\xi_1\big)\cosh\big(\tfrac{3}{2}\sqrt{s}\,\xi_2\big)\Big]^{-1},$$

since the two columns $(\sqrt{2/3}, -1/\sqrt{6}, -1/\sqrt{6})$ and $(0, 1/\sqrt{2}, -1/\sqrt{2})$ are an orthonormal basis of the plane $\mathbf{1}^{\perp}$. For $p = F(z)$ and a direction h with $R = \max_i h_i - \min_i h_i$,

$$D^2 F_1(z)[h,h] = p_1\big((h_1 - \textstyle\sum_j p_j h_j)^2 - \mathrm{Var}_p(h)\big).$$

The first term lies between 0 and $p_1(1-p_1)^2 R^2 \le \frac{4}{27} R^2$, because $\frac{4}{27} - p(1-p)^2 = (p - \frac{1}{3})^2(\frac{4}{3} - p)$, and the variance lies between 0 and $R^2/4$. Hence $|D^2 F_1[h,h]| \le R^2/4 \le |h|^2/2$, and the Hessian of f_s has operator norm at most $\frac{9s}{4}$ everywhere.

Divide $[-6,6]$ into 96 intervals of length $\frac{1}{8}$. For an interval I_j let w_j be its Gaussian mass, m_j the conditional mean and v_j the mass times the conditional variance, and put $P = \sum_j w_j$, $V = \sum_j v_j$. On a product cell $I_j \times I_k$ expand f_s about (m_j, m_k): the linear term integrates to zero and the remainder is at most $\frac{9s}{8}|\xi - (m_j, m_k)|^2$. Summing over the cells, whose integrated squared distances add up to $2PV$, and using $0 \le f_s \le 1$ off the square,

$$g_G(s) \le \sum_{j,k} w_j w_k f_s(m_j, m_k) + \tfrac{9s}{4} PV + 1 - P^2 - \tfrac{1}{3}. \tag{6}$$

Let $t_i = 10/(100 - i)$ for $0 \le i \le 85$, so that $t_0 = \frac{1}{10}$ and $t_{85} = \frac{2}{3}$. In Section 7 the right side of (6) at $s = t_{i+1}$ is enclosed by a rational number U_i, and exact arithmetic gives

$$t_i - U_i > \tfrac{9}{20} t_{i+1}^2, \qquad t_i - U_i > \tfrac{13}{200} t_{i+1} \qquad (0 \le i < 85).$$

By (5) with $l = t_i$ and $r = t_{i+1}$, both bounds hold on $[t_i, t_{i+1}]$. □

5.2. **One replacement.** Let μ be a symmetric posterior law with moments M, U, T as in Section 3, and $a = 3p - \mathbf{1}$, $A = |a|^2$. Given the colour of the parent, the logarithmic likelihood vector of one branch through an edge of strength λ is $P \log(1 + \lambda a)$, computed coordinatewise; its Gaussian counterpart is $G \sim N(\lambda^2 M v_i, \frac{\lambda^2 M}{2} P)$ given $\sigma = i$, where $v_i = \frac{1}{2}(e_i - \frac{1}{3}\mathbf{1})$. For a smooth function f on $\mathbb{R}^3$, invariant under adding a constant to every coordinate, suppose that at every point

$$\begin{gathered} |Df[v_i]| \le g_1, \quad |D^2 f[h,h]| \le c_2 |Ph|^2, \quad |D^3 f[h,h,h]| \le c_3 |Ph|^3, \\ |D^3 f[r,h,h]| \le c_{31} |Pr| |Ph|^2, \quad |D^4 f[h,h,h,h]| \le c_4 |Ph|^4. \end{gathered} \tag{7}$$

For $0 < L < 1$ put

$$B_4 = \tfrac{25}{288} + \tfrac{49}{100} \tfrac{L}{1-L}, \qquad B_3 = \tfrac{1}{6} + \tfrac{49}{20} L B_4, \qquad B_1 = \tfrac{5}{24} + \tfrac{49}{20} L B_3,$$

$$\Gamma_U(f;L) = g_1\Big(\tfrac{1}{6} + \tfrac{49}{20} L B_4\Big) + \tfrac{c_2}{2}\Big(\tfrac{1}{24} + B_3\big(\tfrac{1}{1-L} + 1 + \tfrac{L}{2}\big) + \tfrac{1}{6}\Big) + \tfrac{c_{31} B_1}{2(1-L)^2} + \tfrac{c_3}{6} + \tfrac{c_4}{24(1-L)^4},$$

$$\Gamma_M(f;L) = \tfrac{c_2}{12} + \tfrac{5 c_{31}}{24} + \tfrac{5 c_3 L^2}{72} + \tfrac{c_4(2 + 4L^2 + L^4)}{24},$$

and, for $\{i,j,k\} = \{1,2,3\}$,

$$\ell_i^f = \tfrac{1}{4}\Big(\textstyle\sum_r D_r^3 f - D_i f - (D_j - D_k)^2 f\Big). \tag{8}$$

Here $D_r = \partial/\partial z_r$, so that $D_r^3 f = D^3 f[e_r, e_r, e_r]$ and $(D_j - D_k)^2 f = D^2 f[e_j - e_k, e_j - e_k]$.

Lemma 5.2. *Let $0 < \lambda \le L < 1$ and let W be any random vector, independent of the branch given σ. Then, given $\sigma = i$,*

$$\mathbb{E} f\big(W + P\log(1 + \lambda a)\big) - \mathbb{E} f(W + G) \le \lambda^3 T\, \mathbb{E}\, \ell_i^f(W) + \lambda^4\big(\Gamma_U(f;L)\, U + \Gamma_M(f;L)\, M^2\big).$$

Proof. Take $i = 1$; the other cases follow by permuting colours. Given $\sigma = 1$, the child's point a has law $(1 + \lambda a_1)\mu(da)$. Squares and cubes of vectors below are coordinatewise. Write

$$\Lambda = P\log(1 + \lambda a) = \lambda a + \lambda^2 b + \lambda^3 c + r_4, \qquad b = -\tfrac{1}{2}Pa^2, \quad c = \tfrac{1}{3}Pa^3.$$

Since $a_1 + a_2 + a_3 = 0$ we have $|b| = A/(2\sqrt{6})$, $c = Aa/6$, $|Pa^4|^2 \le \frac{25}{216}A^4$ and $A \le 6$. The integral form of the remainder in $\log(1+t) = t - \frac{t^2}{2} + \frac{t^3}{3} - \dots$, valid for $t \ge -\lambda > -1$, then gives

$$\begin{gathered} |r_4| \le \lambda^4 A^2 B_4, \qquad |\Lambda - \lambda a - \lambda^2 b| \le \lambda^3 A^{3/2} B_3, \qquad |\Lambda - \lambda a| \le \lambda^2 A B_1, \\ |\Lambda| \le \frac{\lambda\sqrt{A}}{1 - L}, \qquad |\lambda a + \lambda^2 b| \le \lambda\sqrt{A}\big(1 + \tfrac{L}{2}\big). \end{gathered} \tag{9}$$

Let $m = \lambda^2 M v_1$ and $C = \frac{\lambda^2 M}{2}P$, so that $G \sim N(m, C)$. Condition on W and expand $f(W + y)$ to fourth order in y, for $y = \Lambda$ and $y = G$; the fourth-order remainder is at most $c_4|Py|^4/24$. We compare the two expansions term by term. For any equivariant plane-valued function V, $\mathbb{E}_\mu[a_1 V(a)] = v_1 \mathbb{E}_\mu[a \cdot V(a)]$, and every radial quantity has the same mean under μ and under $(1 + \lambda a_1)\mu$.

First order. By the identity just stated, $\mathbb{E}_1\Lambda = m - \frac{1}{2}\lambda^3 T v_1 + \frac{1}{6}\lambda^4 U v_1 + \lambda v_1 \mathbb{E}[a \cdot r_4]$. The term in λ^3 is kept; by (9) and $\sqrt{6} \le \frac{49}{20}$ the others contribute at most $\lambda^4 U g_1(\frac{1}{6} + \frac{49}{20}LB_4)$.

Second order. The difference $\mathbb{E}_1\Lambda\Lambda^{\mathsf{T}} - \mathbb{E}GG^{\mathsf{T}}$ is $-\frac{1}{2}\lambda^3 T(e_2 - e_3)(e_2 - e_3)^{\mathsf{T}}$ plus a matrix of nuclear norm at most $\lambda^4\{U[\frac{1}{24} + B_3(\frac{1}{1-L} + 1 + \frac{L}{2}) + \frac{1}{6}] + \frac{1}{6}M^2\}$. Indeed, $\|zz^{\mathsf{T}} - ww^{\mathsf{T}}\|_* \le |z - w|(|z| + |w|)$ compares Λ with $\lambda a + \lambda^2 b$; the term $\lambda^4 bb^{\mathsf{T}}$ costs $U/24$; the change of density acting on $\lambda^3(ab^{\mathsf{T}} + ba^{\mathsf{T}})$ costs $U/6$; and mm^{T} has norm $\lambda^4 M^2/6$. With the Taylor factor $\frac{1}{2}$ and the bound c_2 on the Hessian this gives the c_2 terms of Γ_U and Γ_M.

Third order. Let $q(y) = D^3 f(W)[y, y, y]$. Along the segment from λa to Λ, $|q(\Lambda) - q(\lambda a)| \le 3c_{31}\lambda^4 A^2 B_1/(1 - L)^2$. The change of density acting on $q(\lambda a)$ is at most $c_3\lambda^4 U$, because $|a_1| \le \sqrt{A}$, and what remains is $\lambda^3 \mathbb{E}_\mu q(a)$, which carries the cubic term. For the Gaussian, with $G = m + H$, $\mathbb{E}q(G) = q(m) + 3\mathbb{E}D^3 f(W)[m, H, H]$; here $\mathbb{E}|H|^2 = \lambda^2 M$, $|m| = \lambda^2 M/\sqrt{6}$, $M \le 6$ and $1/\sqrt{6} \le \frac{5}{12}$, which bound the two terms by $\lambda^4 M^2 \cdot \frac{5}{12}c_3 L^2$ and $\lambda^4 M^2 \cdot \frac{5}{4}c_{31}$. With the Taylor factor $\frac{1}{6}$ this gives the c_3 and c_{31} terms.

Fourth order. By (9), $\mathbb{E}_1|\Lambda|^4 \le \lambda^4 U/(1 - L)^4$, and $\mathbb{E}|G|^4 = \lambda^4 M^2(2 + \frac{2}{3}\lambda^2 M + \frac{1}{36}\lambda^4 M^2) \le \lambda^4 M^2(2 + 4L^2 + L^4)$.

Finally, the terms in λ^3 are $-\frac{1}{4}TD_1 f$ from the mean, $-\frac{1}{4}T(D_2 - D_3)^2 f$ from the second moment, and $\frac{1}{6} \cdot \frac{3}{2}T\sum_r D_r^3 f$ from the third, since $\mathbb{E}_\mu D^3 f[a, a, a] = \frac{3}{2}T\sum_r D_r^3 f$ for a symmetric law. Their sum is $\lambda^3 T\ell_1^f(W)$. □

5.3. **The two comparisons.** Write $Q(z) = |3F(z) - \mathbf{1}|^4$. Applied to $f = F_1$ and to $f = Q$, Lemma 5.2 needs the constants in (7).

Lemma 5.3. *For $f = F_1$ the bounds (7) hold with $(g_1, c_2, c_3, c_{31}, c_4) = (\frac{1}{8}, \frac{1}{5}, \frac{2}{5}, \frac{1}{2}, \frac{3}{5})$, and for $f = Q$ with $(8, 20, 38, 38, 99)$.*

Proof. For F_1, $DF_1[v_i] = \frac{1}{2}p_1(\delta_{1i} - p_i)$, which is at most $\frac{1}{8}$ in absolute value. Both functions are invariant under adding a constant to z, so every derivative sees only Ph, and a zero direction is trivial. For the remaining bounds translate and rescale h so that its smallest coordinate is 0 and its largest is 1. Using $h \mapsto -h$ and the symmetry between colours 2 and 3, it suffices to take $h = (0, t, 1)$ or $h = (t, 0, 1)$ with $0 \le t \le 1$ for F_1, and $h = (0, t, 1)$ for the symmetric function Q; in all cases $|Ph|^2 = \frac{2}{3}(1 - t + t^2) > 0$. Write $p = (u, (1-u)v, (1-u)(1-v))$ with $(u, v) \in [0, 1]^2$, which covers the closed simplex. With $m = \sum_i p_i h_i$, $r = h_1 - m$ and the cumulants k_2, k_3, k_4 of

h under p, differentiation of $p_i' = p_i(h_i - m)$ gives

$$D^2F_1 = p_1(r^2 - k_2), \quad D^3F_1 = p_1(r^3 - 3rk_2 - k_3), \quad D^4F_1 = p_1(r^4 - 6r^2k_2 - 4rk_3 + 3k_2^2 - k_4),$$

and the derivatives of $Q = (9\sum_i p_i^2 - 3)^2$ are obtained in the same way. Each bound is then a polynomial inequality in $(u, v, t) \in [0,1]^3$ with rational coefficients: $\frac{1}{5}|Ph|^2 \pm D^2F_1 \geq 0$, $\frac{4}{25}|Ph|^6 - (D^3F_1)^2 \geq 0$, $\frac{3}{5}|Ph|^4 \pm D^4F_1 \geq 0$, and for Q the inequalities $8 \pm \frac{1}{2}D_1Q \geq 0$, $20|Ph|^2 \pm D^2Q \geq 0$, $38^2|Ph|^6 - (D^3Q)^2 \geq 0$, $99|Ph|^4 \pm D^4Q \geq 0$ and $38^2|Ph|^4 - \sum_j (V_j)^2 \geq 0$, where $V_j = D^3Q[e_j, h, h]$. These eighteen inequalities are proved in Section 7 by Bernstein expansion with exact subdivision. The last one gives the mixed bound for Q, since $V \perp \mathbf{1}$ and $D^3Q[r, h, h] = \sum_j r_j V_j$. For F_1 the mixed bound follows from the cubic one by polarization: for unit vectors r, h in the plane,

$$3D^3F_1[h, h, r] = D^3F_1[(h + \tfrac{r}{2})^3] - D^3F_1[(h - \tfrac{r}{2})^3] - \tfrac{1}{4}D^3F_1[r^3],$$

and $|h + \frac{r}{2}|^3 + |h - \frac{r}{2}|^3 \leq \frac{7}{2}$, so $|D^3F_1[h, h, r]| \leq \frac{1}{3} \cdot \frac{2}{5} \cdot \frac{15}{4} = \frac{1}{2}$. □

The third-order terms have a sign. At a point z with $p = F(z)$, let $e_2 = \sum_{i<j} p_ip_j$ and $e_3 = p_1p_2p_3$, and write $\ell_i^{F_i}$ for (8) applied to $f = F_i$.

Lemma 5.4. *At every $z \in \mathbb{R}^3$, with $\beta = \frac{49}{4}$,*

$$\sum_i p_i \ell_i^{F_i}(z) = -e_2(e_2 - 9e_3) \leq 0, \qquad \sum_i p_i \ell_i^{Q}(z) + 9\beta \sum_i p_i \ell_i^{F_i}(z) \leq 0.$$

Proof. Since $D_jF_i = p_i(\delta_{ij} - p_j)$, both sums are polynomials in p, and direct expansion gives

$$\sum_i p_i \ell_i^{F_i} = -e_2(e_2 - 9e_3), \qquad \sum_i p_i \ell_i^{Q} = 54\big(18e_2^3 - 270e_2^2e_3 - 5e_2^2 + 99e_2e_3 - 6e_3\big).$$

By Maclaurin's inequality $e_2 \geq 3e_3^{2/3}$, and $e_3 \leq \frac{1}{27}$, so $e_2 \geq 9e_3$; this is the first claim.

For the second, put $a = 3p - \mathbf{1}$, $A = |a|^2$ and $J = \sum_i a_i^3$, so that $e_2 = (6 - A)/18$ and $e_3 = (J + 3 - \frac{3}{2}A)/81$. Substitution shows that the left side of the second inequality equals $C(A) + B(A)J$, where

$$C(A) = \big(4 - \tfrac{\beta}{3}\big)A + \big(\tfrac{\beta}{18} - 4\big)A^2 + \tfrac{2}{3}A^3, \qquad B(A) = \tfrac{\beta}{3} - 2 + \big(3 - \tfrac{\beta}{18}\big)A - \tfrac{5}{9}A^2.$$

On the simplex $0 \leq A \leq 6$. Since $a_1 + a_2 + a_3 = 0$, the discriminant identity $A^3/6 - J^2 = \frac{1}{3}\prod_{i<j}(a_i - a_j)^2$ gives $J \leq A^{3/2}/\sqrt{6}$, and $e_3 \geq 0$ gives $J \geq \frac{3}{2}A - 3$. If $B(A) \geq 0$, put $t = \sqrt{A/6} \in [0,1]$ and use $J \leq 6t^3$:

$$C + BJ \leq -2t^2(1-t)^2\Big[48\big(t - \tfrac{23}{384}\big)^2 + \tfrac{239}{3072} + 60t^3\Big] \leq 0.$$

If $B(A) < 0$, use $J \geq \frac{3}{2}A - 3$:

$$C + BJ \leq -\tfrac{1}{144}(6 - A)^2(24A + 25) \leq 0.$$

Both right sides are exact polynomial identities in t and A respectively. □

Proposition 5.5. *Let μ be a symmetric posterior law with $T(\mu) \geq 0$, let $d \geq 1$, $0 < \lambda \leq L < 1$, and let x', U' be the second and fourth moments x and U of $T_\lambda\mu$. With $s = d\lambda^2 x$, $\beta = \frac{49}{4}$, $f_U = \Gamma_U(F_1; L)$, $f_M = \Gamma_M(F_1; L)$, $q_U = \Gamma_U(Q; L)$ and $q_M = \Gamma_M(Q; L)$,*

$$x' - g_G(s) \leq d\lambda^4(f_U U + f_M M^2), \tag{10}$$

$$U' - U_G(s) + 9\beta\big(x' - g_G(s)\big) \leq d\lambda^4\big((q_U + 9\beta f_U)U + (q_M + 9\beta f_M)M^2\big). \tag{11}$$

Proof. Given σ, the logarithmic likelihood of the d branches is the sum of d independent copies of $P\log(1+\lambda a)$, and that of the Gaussian experiment at $s = d\lambda^2 x$ is the sum of d independent copies of G, since $d\lambda^2 M/2 = \frac{9s}{2}$. Replace the branches by Gaussian copies one at a time. Every intermediate sum is the logarithmic likelihood of a genuine symmetric experiment, since sums of independent likelihoods remain likelihoods, so its posterior is $F(W)$ and its law ν satisfies $\mathbb{P}(W \in \cdot \mid \sigma = i) = 3F_i(W)\,\nu$. Hence the third-order term of Lemma 5.2 for $f = F_\sigma$, averaged over σ, is $\lambda^3 T$ times the ν-average of $\sum_i p_i \ell_i^{F_i}$, and for $f = Q$ it is $\lambda^3 T$ times the ν-average of $\sum_i p_i \ell_i^{Q}$. By calibration $\mathbb{E}F_\sigma(\cdot) = x + \frac{1}{3}$ at both ends of the chain, with $x = x'$ for the branches and $x = g_G(s)$ for the Gaussian, and Q is symmetric, so its conditional and unconditional means agree. Summing the d replacements, applying Lemma 5.2 to F_σ for (10) and to Q and $9\beta F_\sigma$ separately for (11), the third-order terms are nonpositive by Lemma 5.4 and $T \geq 0$, and may be dropped. □

5.4. **Exclusion of large fixed points.**

Proposition 5.6. *Let $d \geq 91$ and $\lambda = d^{-1/2}$. Every symmetric posterior law μ with $T_\lambda \mu = \mu$ and $T(\mu) \geq 0$ has $x(\mu) < \frac{1}{10}$.*

Proof. Put $D = 91$, $L = 209657/2000000$ and $\beta = \frac{49}{4}$. Then $DL^2 \geq 1$, so $\lambda \leq L$, and Proposition 5.5 applies. Let f_U, f_M, q_U, q_M be as there, at this L, and define

$$E_d(s) = \frac{d - q_U}{f_U}\big(s - g_G(s)\big) - 81\Big[\Big(1 - \frac{q_U}{d}\Big)\frac{f_M}{f_U} + \frac{q_M}{d}\Big]s^2 - U_G(s).$$

In Section 7 we verify, in exact rational arithmetic,

$$D - q_U - 9\beta f_U > 0, \qquad q_M - q_U\frac{f_M}{f_U} > 0, \qquad E_D(s) > \tfrac{3}{200} \quad \big(\tfrac{1}{10} \leq s \leq \tfrac{2}{3}\big). \tag{12}$$

Suppose $x = x(\mu) \geq \frac{1}{10}$; then $x \leq \frac{2}{3}$. At the fixed point $x' = x$, $U' = U$, $s = d\lambda^2 x = x$ and $d\lambda^4 = 1/d$. Write $M = 9x$, $K = U/(81x^2)$ and $\gamma = (x - g_G(x))/x^2$. Then (10) reads $K \geq (d\gamma/81 - f_M)/f_U$, and (11) reads

$$\big(d - q_U - 9\beta f_U\big)K \leq \frac{dU_G(x)}{81x^2} - \frac{\beta d\gamma}{9} + q_M + 9\beta f_M.$$

The coefficient on the left is positive by (12), since $d \geq D$. Inserting the lower bound for K, the terms in β cancel, and after multiplication by $81x^2/d$ what remains is $E_d(x) \leq 0$.

By Lemma 5.1, $x - g_G(x) > 0$. The coefficient $(d - q_U)/f_U$ increases with d, while $81[(1 - q_U/d)f_M/f_U + q_M/d] = 81[f_M/f_U + (q_M - q_U f_M/f_U)/d]$ decreases with d by (12). Hence $E_d(x) \geq E_D(x) > \frac{3}{200}$, a contradiction. □

Remark 5.7. Without the second observable, (10) alone does not bound U, and the third-order term of Q by itself has no sign: $\sum_i p_i \ell_i^Q > 0$ at $p = (\frac{2}{5}, \frac{3}{10}, \frac{3}{10})$. The multiple $9\beta F_\sigma$ of the first observable corrects this sign everywhere on the simplex (Lemma 5.4), and its contribution cancels in the proof of Proposition 5.6. The value $\beta = \frac{49}{4}$ is one convenient choice; the same proof works for any β for which $C(A) + B(A)J \leq 0$ on the feasible region and (12) holds.

6. Proof of Theorem 1.1

By Lemma 2.4 with $D = d$, it suffices to take $\lambda = d^{-1/2}$. The ingredients for each degree are collected in Table 2.

Proof of Theorem 1.1. Let $\lambda = d^{-1/2}$.

TABLE 2. The two estimates used at each degree.

degrees	exclusion near δ_u	entry or exclusion away from δ_u
$d = 2$	$x \le r_2 = \frac{23}{2000}$, Proposition 3.4	$x_{500} \le 0.011415$, Proposition 4.4
$d = 3$	$x \le r_3 = \frac{3}{100}$, Proposition 3.6	$x_{106} \le 0.018465$, Proposition 4.4
$4 \le d \le 90$	$x \le r_d$ in (2), Proposition 3.6	Table 1, Proposition 4.4
$d \ge 91$	$x \le r_d$, $r_d \ge \frac{179}{1500}$, Proposition 3.6	$x(\mu) < \frac{1}{10}$ if $T(\mu) \ge 0$, Proposition 5.6

Degrees $2 \le d \le 90$. By Proposition 4.4 there is N with $x_N < r_d$. By Proposition 3.4 for $d = 2$ and Proposition 3.6 for $d \ge 3$, every symmetric posterior law μ with $T_\lambda \mu = \mu$ and $x(\mu) \le r_d$ equals δ_u. Proposition 2.3 with $r = r_d$ gives nonreconstruction.

Degrees $d \ge 91$. By Lemma 2.2, μ_∞ is a symmetric posterior law with $T_\lambda \mu_\infty = \mu_\infty$. Since $d\lambda^3 = d^{-1/2} < 1$, Corollary 3.3 gives $T(\mu_\infty) \ge 0$, and Proposition 5.6 gives $x(\mu_\infty) < \frac{1}{10}$. For $d \ge 80$ we have $r_d \ge \varrho_d = \frac{179}{1500} > \frac{1}{10}$, so $\mu_\infty = \delta_u$ by Proposition 3.6. Hence $x_n \to 0$, and nonreconstruction follows as in the proof of Proposition 2.3.

Finally, for $\lambda < d^{-1/2}$, Lemma 2.4 gives $x_n(d, \lambda) \le x_n(d, d^{-1/2}) \to 0$. Since reconstruction holds for $\lambda > d^{-1/2}$ by [5], the threshold is $d^{-1/2}$. □

7. THE COMPUTATIONS

Every inequality used above that is not proved by hand is a finite list of inequalities between rational numbers. This section describes how each list is produced and checked. The programs are in the accompanying repository; each check has a producer and a separate replay that does not import the producer.

7.1. **Bernstein certificates.** Lemma 3.2 and Propositions 3.4 and 3.6 and Lemma 5.3 reduce to the nonnegativity of polynomials with rational coefficients on a box, which after an affine change of variables is $[0,1]^k$ with $2 \le k \le 4$. Where a polynomial vanishes on part of the boundary, the monomial factors responsible are divided out first, exactly, and the quotient is treated. For a polynomial $\sum_e a_e x^e$ of degree n_j in x_j the Bernstein coefficients on the box are

$$b_\kappa = \sum_{e \le \kappa} a_e \prod_j \binom{\kappa_j}{e_j} \Big/ \binom{n_j}{e_j},$$

and the polynomial is a convex combination of them at every point of the box. If some coefficient is negative, the box is bisected, the coordinates taken in turn, and the coefficients on the two halves are obtained exactly by de Casteljau's algorithm. A certificate is a prefix-free set of dyadic subboxes of total volume one, each with nonnegative coefficients; a negative coefficient is never taken as a counterexample. All arithmetic is in integers with a common denominator. For Lemma 5.3 the ten inequalities for F_1 close with 68 leaves and the eight for Q with 124 leaves. Exact negative controls show that the constants cannot be lowered to $\frac{1}{10}$, $\frac{1}{5}$ and $\frac{2}{5}$ in place of $\frac{1}{5}$, $\frac{2}{5}$ and $\frac{3}{5}$: at $p = (\frac{1}{4}, 0, \frac{3}{4})$ and $h = (0, \frac{1}{2}, 1)$ one has $|Ph|^2 = \frac{1}{2}$, $|D^2 F_1| = \frac{3}{32}$ and $|D^4 F_1| = \frac{15}{128}$, and at $p = (\frac{1}{2}, 0, \frac{1}{2})$ with the same h, $|D^3 F_1| = \frac{1}{8}$.

7.2. **Entry bounds.** The laws ν_n of Section 4 are computed with integer masses, as described there, from the parameters in Table 1. The output of each row is the exact rational number $x(\nu_{n_0})$, stored as a fraction; the comparison with r_d is made in exact arithmetic for every d in the row, by squaring where r_d involves $\sqrt{d}$. The computation is deterministic, and a replay recomputes each row from the parameters alone.

7.3. **Gaussian quadrature.** Lemma 5.1 and the third inequality in (12) need upper bounds for g_G and U_G at the points $t_i = 10/(100-i)$, $0 \le i \le 85$. For Lemma 5.1 they are obtained from (6). For (12) the same quadrature is used with 192 intervals of length $\frac{1}{16}$ on $[-6,6]$, for F_1 and for Q; for Q the cell error is at most $90s\,PV$, since $|D^2Q[h,h]| \le 20|Ph|^2$ and the map from ξ to the logits has squared norm $\frac{9s}{2}$, and the tail is at most $36(1-P^2)$, since $0 \le Q \le 36$. For F_1 the same argument with $c_2 = \frac{1}{5}$ gives the cell error $\frac{9s}{10}PV$. The masses w_j, the moments m_j and v_j, the exponentials and the sums are evaluated in ball arithmetic with 160 bits [4], with outward rounding throughout, and the upper endpoints of the resulting balls are exported as exact dyadic rationals. All further comparisons use these rationals. A replay recomputes the endpoint bounds from the stored cell data with independent code and, at t_0, t_{42} and t_{85}, evaluates the quadrature sums afresh on all 192^2 cells and checks that the two enclosures overlap.

7.4. **The large-degree certificate.** At $L = \frac{209657}{2000000}$ the constants of Proposition 5.5 are

$$f_U = 0.27741\ldots, \quad f_M = 0.17224\ldots, \quad q_U = 27.0695\ldots, \quad q_M = 18.0441\ldots,$$

all exact rationals. Then $D - q_U - 9\beta f_U = 33.346\ldots$ and $q_M - q_U f_M/f_U = 1.2367\ldots$, which are the first two inequalities of (12). For the third write $E_D(s) = \alpha s - \alpha g_G(s) - Bs^2 - U_G(s)$ with $\alpha = (D-q_U)/f_U = 230.458\ldots$ and $B = 81[(1-q_U/D)f_M/f_U + q_M/D] = 51.393\ldots$. The upper bounds for g_G and U_G give lower bounds for E_D at the 86 points t_i, and the following lemma controls E_D between them.

Lemma 7.1. *For $s > 0$, $|g_G''(s)| \le \frac{1269}{40} < 32$ and $|U_G''(s)| \le \frac{14229}{4} < 3600$.*

Proof. Given $\sigma = 1$, the vector $Z(s)$ is a Brownian motion in the plane $\mathbf{1}^\perp$ with drift $m_0 = (3, -\frac{3}{2}, -\frac{3}{2})$ and covariance $\frac{9}{2}P$ per unit time, so for a smooth function f with bounded derivatives $\frac{d^2}{ds^2}\mathbb{E}f(Z(s)) = \mathbb{E}\,\mathcal{L}^2 f(Z(s))$ with $\mathcal{L} = m_0 \cdot \nabla + \frac{9}{4}\Delta_P$. Let e, e' be an orthonormal basis of the plane. For a symmetric 4-tensor with $|D^4 f[h^4]| \le c_4|h|^4$, the identity $12D^4f[e,e,e',e'] = D^4f[(e+e')^4] + D^4f[(e-e')^4] - 2D^4f[e^4] - 2D^4f[e'^4]$ gives $|D^4f[e,e,e',e']| \le c_4$, and so $|\Delta_P^2 f| \le 4c_4$. With $|m_0|^2 = \frac{27}{2}$ and $|m_0| < \frac{15}{4}$,

$$|\mathcal{L}^2 f| \le \tfrac{27}{2}c_2 + 2 \cdot \tfrac{9}{4} \cdot 2c_{31}|m_0| + \tfrac{81}{16} \cdot 4c_4 \le \tfrac{27}{2}c_2 + \tfrac{135}{4}c_{31} + \tfrac{81}{4}c_4.$$

The constants of Lemma 5.3 give $\frac{1269}{40}$ for $f = F_1$ and $\frac{14229}{4}$ for $f = Q$. □

Hence $|E_D''| \le C_D = 32\alpha + 2B + 3600 = 11077.4\ldots$, and on each interval $[t_i, t_{i+1}]$

$$E_D(s) \ge \min\{E_D(t_i), E_D(t_{i+1})\} - \tfrac{1}{8}C_D(t_{i+1} - t_i)^2.$$

For all 85 intervals the right side, computed from the exported lower bounds, exceeds $0.01584 > \frac{3}{200}$. Two controls are part of the check: raising the bound for U_G by one at every point makes it fail, and so does removing one endpoint.

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Declaration of competing interest

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Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. In October 2026 OpenAI GPT models, running as research tasks of that project, wrote the proofs and the programs that produce the computer-assisted steps, and a second OpenAI GPT model audited each piece. Claude models (Anthropic), working in Claude Code, directed the work across degrees, audited every piece again with programs of their own, assembled the argument, and wrote the paper. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

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